

Solutions for Spl Lab Q-2024

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B.Sc (Hons.) in Computer Science and Engineering

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Course Code 510202 (According to new syllabus)

Course Title: Structured Programming Language LAB

Time: 03 Hours Full Marks: 40

[There are two sections in this question. Implement any one from each of the following sections.]

Section-A

1. Write a C program to find area and circumference of a circle (use symbolic constant).
2. Write a C program to find maximum and minimum among three numbers.
3. Write a C program to verify whether the given number is odd or even?
4. Write a C program to verify whether the given year is a leap year or not?
5. Write a C program to find all possible roots of $ax^2+bx+c=0$.
6. Write a C program to find the series of all prime numbers up to value 100.
7. Write a C program to find the factorial of an integer.
8. Write a C program to check whether the given number is an Armstrong number or not?
9. Write a C program to find GCD of two integers.
10. Write a C program that takes an integer as input and display it in reverse order.

Section-B

11. Write a C program to display all elements of an array and also calculate sum of elements of that array.
12. Write a C program to search an element from a linear array.
13. Write a C program to find sum of two matrices of size 3×3 .
14. Write a C program to exchange the value of two variables (X,Y) using pointer.
15. Write a C program to perform all basic string handling operations.
16. Write a C program to determine whether a string is palindrome or not.
17. Write a C program to print n Fibonacci numbers using iterative approach.
18. Write a C program to perform bubble sort on 10 integer numbers in an ascending order.
19. Write a C program to record student's information according to their merit position.
20. Write a C program to read data from keyboard, write it to a file called INPUT.txt. Again open the same file to read the same data from the file and display it to the terminal.

Marks Distribution:

i) Algorithm	:	5x2=10
ii) Source Code	:	5x2=10
iii) Implementation	:	5x2=10
iv) Viva-Voce	:	5x2=10
Total	:	40

1. Write a C program to find area and circumference of a circle (use symbolic constant).

```
#include <stdio.h>

#define PI 3.14

int main(){

    float r, area, cirs;

    printf("Enter radius:");

    scanf("%f", &r);

    cirs = 2 * PI * r;

    area = PI * r * r;

    printf("The area of circle is: %.2f and circumference is: %.2f", area, cirs);

    return 0;

}
```

2. Write a C program to find maximum and minimum among three numbers.

```
#include<stdio.h>
int main(){

int  a ,b , c ,min, max;

printf("Enter three numbers:");
scanf("%d %d %d",&a ,&b ,& c );

//find min

if(a <b && a<c ){
    min = a;
}
else if(b<a && b<c){
    min = b;
}
else{
    min = c;
}
printf("min= %d",min);

//find max

if(a >b && a>c ){
    min = a;
}
else if(b>a && b>c){
    min = b;
}
else{
    min = c;
}

printf("max= %d",max);

return 0;

}
```

3. Write a C program to verify whether the given number is odd or even?

```
#include <stdio.h>int main(){  
  
    int x;  
  
    printf("Enter any number: ");  
    scanf("%d", &x);  
  
    if (x % 2 == 0){  
        printf("%d is even", x);  
    }  
  
    else{  
        printf("%d is odd", x);  
    }  
  
    return 0;  
  
}
```

4. Write a C program to verify whether the given year is a leap year or not?

```
#include <stdio.h>

int main (){

    int year;

    printf("Enter year : \n");

    scanf("%d", &year);

    if( year%400==0 || (year%4==0 && year!=100)){

        printf("Leap year.");

    }

    else {

        printf("Not a leap year.");

    }

    return 0;

}
```

5. Write a C program to find all possible roots of $ax^2+bx+c=0$.

```
#include <stdio.h>

#include <math.h>

int main(){

    int a=0, b, c;

    float x1, x2, d;

    while(a==0){

        printf("Enter value of a: ");

        scanf("%d", &a);

    }

    printf("Enter value of b and c: ");

    scanf("%d%d", &b, &c);

    d = (b * b) - (4 * a * c);

    if (d == 0){ // test case 2 , 4,2

        x1 = -b / (2 * a);

        printf("unique solution x= %.3f", x1);

    }

    else if (d > 0){ // test case 1 , 4,2

        x1 = (-b + sqrt(d)) / (2 * a);

        x2 = (-b - sqrt(d)) / (2 * a);

        printf("x1 = %.3f\nx2 = %.3f", x1, x2);

    }

}
```

```
}  
  
else{ // test case 3 , 4,2  
  
    d *= (-1);  
  
    float m = -b / (2 * a), n = sqrt(d) / (2 * a);  
  
    printf("imaginary solution\n");  
  
    printf("x1 = %.2f + i%.3f\nx2 = %.3f - i%.3f", m, n, m, n); // Here i means complex  
number  
  
}  
  
return 0;  
  
}
```

6. Write a C program to find the series of all prime numbers up to value 100.

```
#include<stdio.h>
```

```
int main(){
```

```
    for(int i = 2; i<=100; i++ ){
```

```
        int isPrime =1;
```

```
        for(int j=2; j*j<=i; j++){
```

```
            if(i%j==0){
```

```
                isPrime = 0;
```

```
                break;
```

```
            }
```

```
        }
```

```
        if(isPrime){
```

```
            printf("%d ", i);
```

```
        }
```

```
    }
```

```
    return 0;
```

```
}
```


7. Write a C program to find the factorial of an integer.

```
#include <stdio.h>

int main(){

    int n, fact=1;

    printf("Enter The Number: ");

    scanf("%d", &n);

    for(int i=1; i<=n; i++){

        fact *= i;

    }

    printf("Factorial of %d is %d\n", n, fact);

    return 0;

}
```

8. Write a C program to check whether the given number is an Armstrong number or not?

```
// test case 153 , 370
```

```
#include<stdio.h>
```

```
#include<math.h>
```

```
int main(){
```

```
    int number, originalNumber, remainder, result = 0, digit = 0;
```

```
    printf("Enter an integer: ");
```

```
    scanf("%d", &number);
```

```
    originalNumber = number;
```

```
    while (originalNumber != 0) {
```

```
        originalNumber /= 10;
```

```
        ++digit;
```

```
    }
```

```
    originalNumber = number;
```

```
    while (originalNumber != 0) {
```

```
        remainder = originalNumber % 10;
```

```
        result += pow(remainder, digit);
```

```
        originalNumber /= 10;
```

```
}
```

```
if (result == number) {
```

```
    printf("%d is an Armstrong number.\n", number);
```

```
} else {
```

```
    printf("%d is not an Armstrong number.\n", number);
```

```
}
```

```
return 0;
```

```
}
```

9. Write a C program to find GCD of two integers.9. Write a C program to find GCD of two integers.

```
#include <stdio.h>
```

```
int gcd(int a, int b){
```

```
    if(a%b==0) return b;
```

```
    return gcd(b, (a%b));
```

```
}
```

```
int main(){
```

```
    int a,b;
```

```
    scanf("%d %d", &a,&b);
```

```
    printf("gcd is: %d", gcd(a,b));
```

```
    return 0;
```

```
}
```

10. Write a C program that takes an integer as input and display it in reverse order.

```
#include <stdio.h>
```

```
int main(){
```

```
    int n, sum=0;
```

```
    printf("Enter an integers: ");
```

```
    scanf("%d", &n);
```

```
    for(int i=1; n!=0; i++){
```

```
        printf("%d", n%10);
```

```
        n = n/10;
```

```
    }
```

```
    return 0;
```

```
}
```

11. Write a C program to display all elements of an array and also calculate sum of elements of that array.

```
#include <stdio.h>
```

```
int main(){
```

```
    int n ,i, sum=0;
```

```
    scanf("%d",&n);
```

```
    int arr[n];
```

```
    for(i =0 ;i<n;i++){
```

```
        scanf("%d",&arr[i]);
```

```
    }
```

```
    for(i =0 ;i<n;i++){
```

```
        printf("%d ",arr[i]);
```

```
    }
```

```
    for(i=0; i<n; i++){
```

```
        sum += arr[i];
```

```
    }
```

```
    printf("\nSum of all elements is : ", sum);
```

```
    return 0;
```

```
}
```

12. Write a C program to search an element from a linear array.

```
#include<stdio.h>

int main(){

    int n,f;

    printf("Enter the size of array :");

    scanf("%d",&n);

    int arr[n];

    printf("Enter elements :");

    for(int i=0;i<n;i++){

        scanf("%d",&arr[i]);

    }

    printf("Enter the element you want to find :");

    scanf("%d",&f);

    for(int i=0;i<n;i++){

        if(f==arr[i]){

            printf("Element %d found at index : %d \n",f,i);

            return 0;

        }

    }

    printf("Not found in array");

    return 0;

}
```

13. Write a C program to find sum of two matrices of size 3x3.

```
#include <stdio.h>
```

```
int main(){
```

```
int i ,j;
```

```
int a[3][3] ;
```

```
int b[3][3] ;
```

```
int c[3][3] ;
```

```
printf("Enter 1st matrix elements: \n");
```

```
for(i = 0;i<3;i++){
```

```
    for(j = 0;j<3;j++){
```

```
        scanf("%d",&a[i][j]);
```

```
    }
```

```
}
```

```
printf("Enter 2nd matrix elements: \n");
```

```
for(i = 0;i<3;i++){
```

```
    for(j = 0;j<3;j++){
```

```
        scanf("%d",&b[i][j]);
```

```
    }
```



```
}
```

```
// matrix addition
```

```
for(i = 0;i<3;i++){
```

```
    for(j = 0;j<3;j++){
```

```
        c[i][j] = a[i][j] + b[i][j];
```

```
    }
```

```
}
```

```
printf("Result matrix: \n");
```

```
for(i = 0;i<3;i++){
```

```
    for(j = 0;j<3;j++){
```

```
        printf("%d ",c[i][j]);
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
return 0;
```

```
}
```

14. Write a C program to exchange the value of two variables (X,Y) using pointer.

```
#include <stdio.h>
```

```
int main() {
```

```
    int x, y, temp;
```

```
    int *p1, *p2;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &x, &y);
```

```
    p1 = &x;
```

```
    p2 = &y;
```

```
    printf("Before : x = %d, y = %d\n", x, y);
```

```
    temp = *p1;
```

```
    *p1 = *p2;
```

```
    *p2 = temp;
```

```
    printf("After : x = %d, y = %d\n", x, y);
```

```
    return 0;
```

```
}
```

15. Write a C program to perform all basic string handling operations

```
#include<stdio.h>

#include<string.h>

int main(){

    char str1[50],str2[50],str3[50];

    printf("Enter the first string :");

    scanf("%s",str1);

    printf("Enter the second string :");

    scanf("%s",str2);

    printf("length of str1 : %d \n",strlen(str1));

    printf("length of str2 : %d \n",strlen(str2));


    strcpy(str3,str1);

    printf("copied str1 into str3 : %s\n",str3);


    strcat(str3,str2);

    printf("concatenated str2 into str3 :%s \n",str3);


    int cmp =strcmp(str1,str2);

    if(cmp==0){

        printf("str1 and str2 are equal");

    }

    else{

        printf("str1 is greater than str2.");

    }

}
```

```
}
```

```
return 0;
```

```
}
```

16. Write a C program to determine whether a string is palindrome or not.

```
#include <stdio.h>

#include <string.h>

int main() {

    char str[100];

    int len, isPalindrome = 1;

    printf("Enter a string: ");

    scanf("%s", str);


    len = strlen(str);

    for(int i = 0; i < len / 2; i++) {

        if(str[i] != str[len - i - 1]) {

            isPalindrome = 0;

            break;

        }

    }

    if(isPalindrome){

        printf("palindrome");

    }

    else{

        printf("not palindrome");

    }

    return 0;

}
```

17. Write a C program to print n Fibonacci numbers using iterative approach.

```
#include <stdio.h>

int main() {

    int n, x= 0, y =1, sum=1;

    printf("Enter the number of terms : ");

    scanf("%d", &n);

    printf("%d %d", x, y);

    for(int i = 2; i < n; i++){

        sum = x + y;

        x = y;

        y = sum;

        printf(" %d", sum);

    }

    return 0;

}
```

18. Write a C program to perform bubble sort on 10 integer numbers in an ascending order.

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[10];
```

```
    for(int i=0; i<10; i++){
```

```
        printf("enter array element %d ", i+1);
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    for (int i = 0; i < 10; i++) {
```

```
        for (int j = 0; j < 10 - i - 1 ; j++) {
```

```
            if (arr[j] > arr[j + 1]) {
```

```
                // Swap arr[j] and arr[j + 1]
```

```
                int temp = arr[j];
```

```
                arr[j] = arr[j + 1];
```

```
                arr[j + 1] = temp;
```

```
            }
```

```
        }
```

```
    }
```

```
    for(int i=0; i<10; i++){
```

```
    printf("%d ", arr[i]);  
}  
return 0;  
}
```


19. Write a C program to record student's information according to their merit position

```
#include <stdio.h>

typedef struct {
    char name[50];
    int roll;
    float marks;
} Student;

int main() {
    int n, i;

    printf("Enter number of students: ");
    scanf("%d", &n);

    Student students[n];

    for (i = 0; i < n; i++) {
        printf("Enter name, roll number and marks for student %d:\n", i + 1);
        printf("Name: ");
        scanf("%s", students[i].name);
        printf("Roll Number: ");
        scanf("%d", &students[i].roll);
        printf("Marks: ");
        scanf("%f", &students[i].marks);
    }

    // Sort students based on marks in descending order
    for (i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
```

```
        if (students[i].marks < students[j].marks) {  
            Student temp = students[i];  
            students[i] = students[j];  
            students[j] = temp;  
        }  
    }  
}  
  
printf("\nStudent record according to merit:\n");  
  
for (i = 0; i < n; i++) {  
    printf("Merit: %d, Name: %s, Roll Number: %d, Marks: %.2f\n", i+1, students[i].name,  
students[i].roll, students[i].marks);  
}  
  
return 0;  
}
```

20. Write a C program to read data from keyboard, write it to a file called INPUT.txt. Again open the same file to read the same data from the file and display it to the terminal.

```
#include<stdio.h>

int main(){

    FILE *fp;

    char data[1000];

    printf("Enter some text: ");
    fgets(data,sizeof(data),stdin);

    fp = fopen("INPUT.txt","w");

    fputs(data,fp);

    fclose(fp);

    fp = fopen("INPUT.txt","r");

    printf("Text from INPUT file: \n");

    while(fgets(data,sizeof(data),fp) != NULL){

        printf("%s",data);

    }

    fclose(fp);

    return 0;

}
```